ECO 331

Economic History

`The Industrial Revolution`

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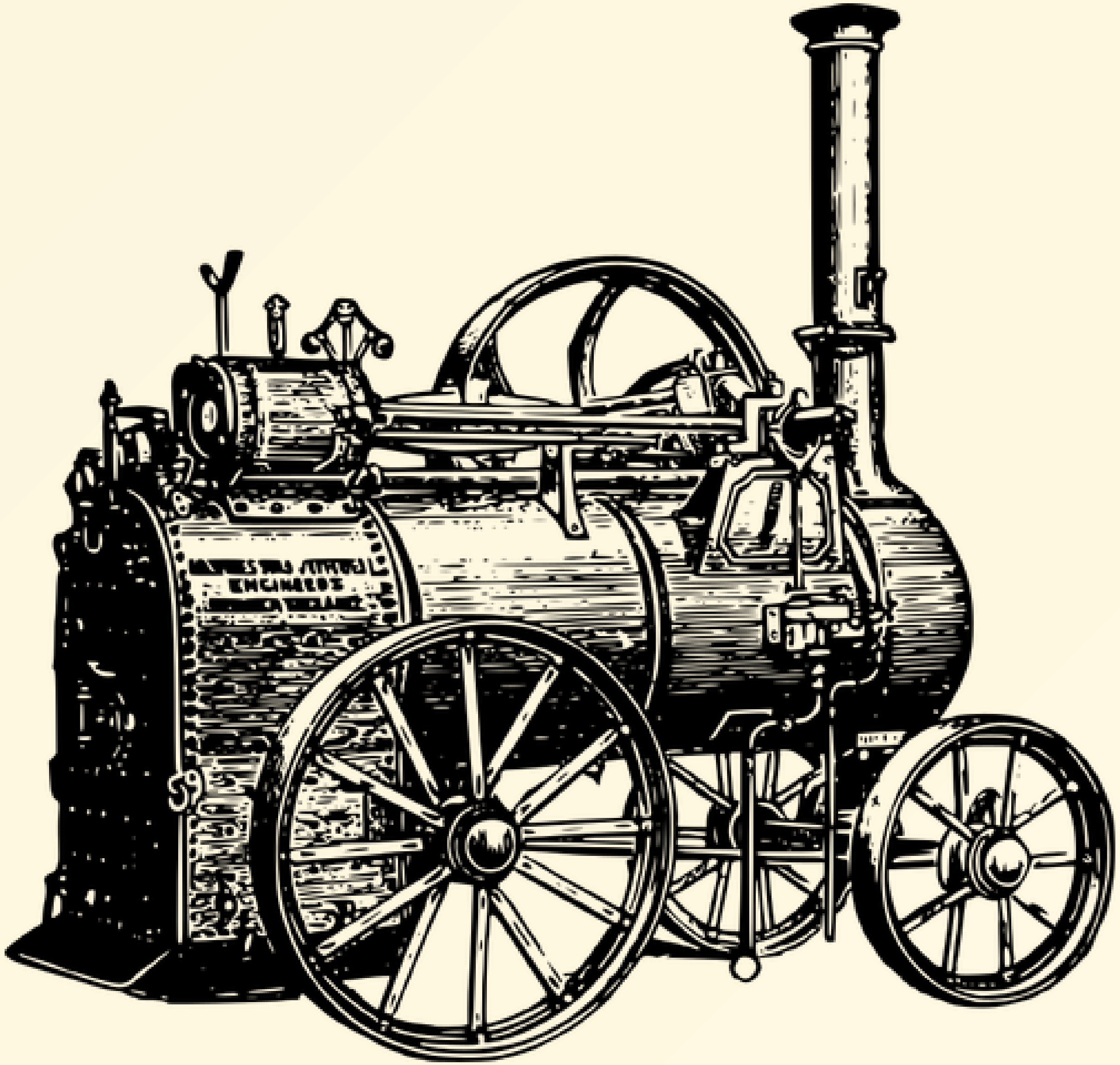
Spring 2026



read by Tue 4/14

- Oded Galor, Ch 4, "Full Steam"
- KR 8 "Britain's Industrial Revolution"

read by Thu 4/16

- Mokyr, Joel, 1992. The Lever of Riches: Technological Creativity and Economic Progress, Oxford. Read chapter 5, "The Years of Miracles: The Industrial Revolution, 1750-1830." (PDF)

Red by Tue 4/21

- Solow, Barbara, 1993. "Slavery and Colonization," Chapter 1 in Slavery and the Rise of the Atlantic System. Cambridge University Press.
- Domar, Evsey D. 1970. “The Causes of Slavery or Serfdom: A Hypothesis.” The Journal of Economic History 30 (1): 18–32. PDF

Britain's Industrial Revolution

- 1700-1870, Britain's GDP grows 10X
- Popn 4X, GDP/capita 4X
- Unprecedented transformation.
- Growth was sustained.

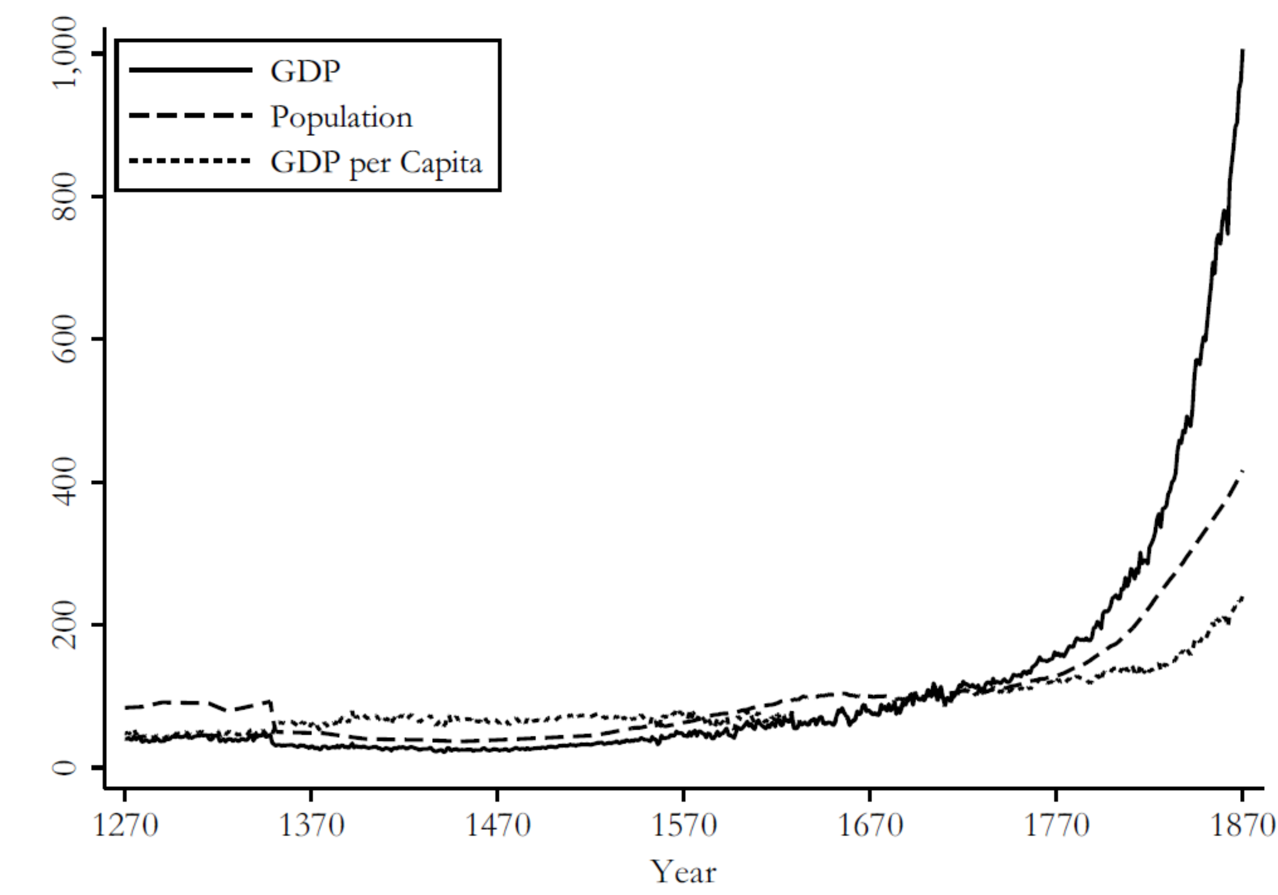
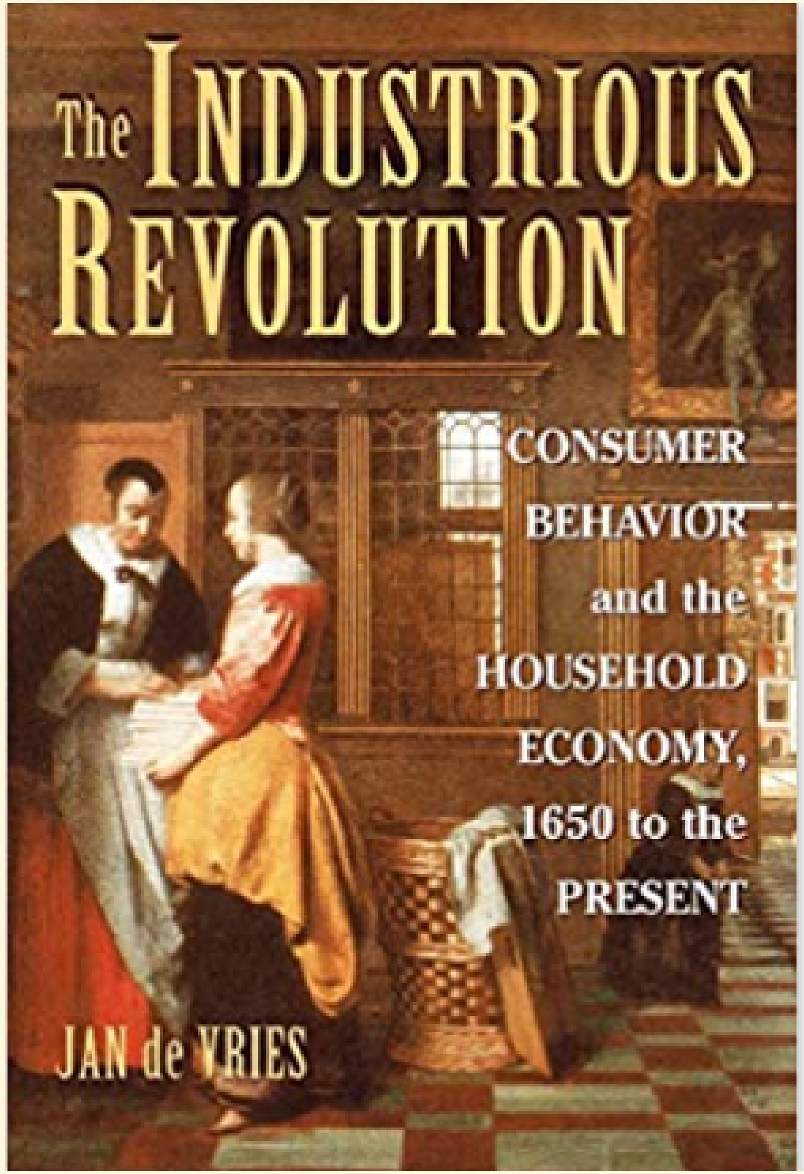


Figure 8.1 GDP, GDP per capita, and population in England/Britain, 1270–1870 (1700 = 100)

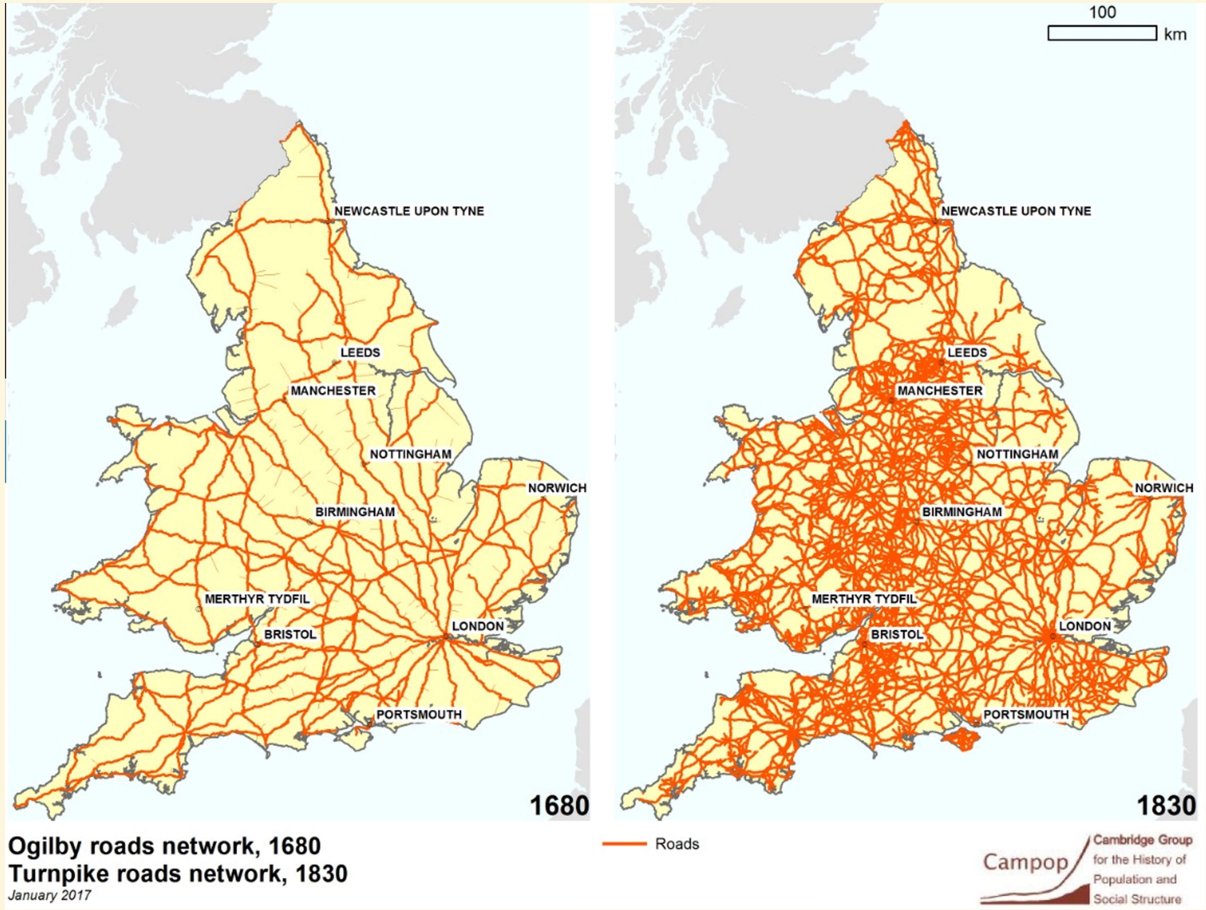
Preconditions

- Commercialization
  - No warfare on British territory since civil war.
  - Commercialization, market integration.
  - New goods: ready-to-wear clothing, tea, sugar, tobacco, spices.
  - De Vries (2008) argues consumer "Industrious Revolution" as households supply more labor to buy new goods (contrast to Marxian interpretation).



The Rise of London

- In 1550: 50-70K (~2% of popn)
  - 2X 2nd largest Norwich.
- By 1700: ~ 575K (10% of popn)
  - 19X Norwich



Transport Improvements

- 18th C transport revolution
- Stagecoach speeds from 1.96 MPH in 1700 to 7.96 in 1820
- Improved road network, stagecoach design, way-stations
- 1840: Britain 2X road density of France
- Canals link NW industrial heartland w/ coal
- Rail travel in 1870 10x faster than 1700 coach
- Map of turnpikes 1680-1830

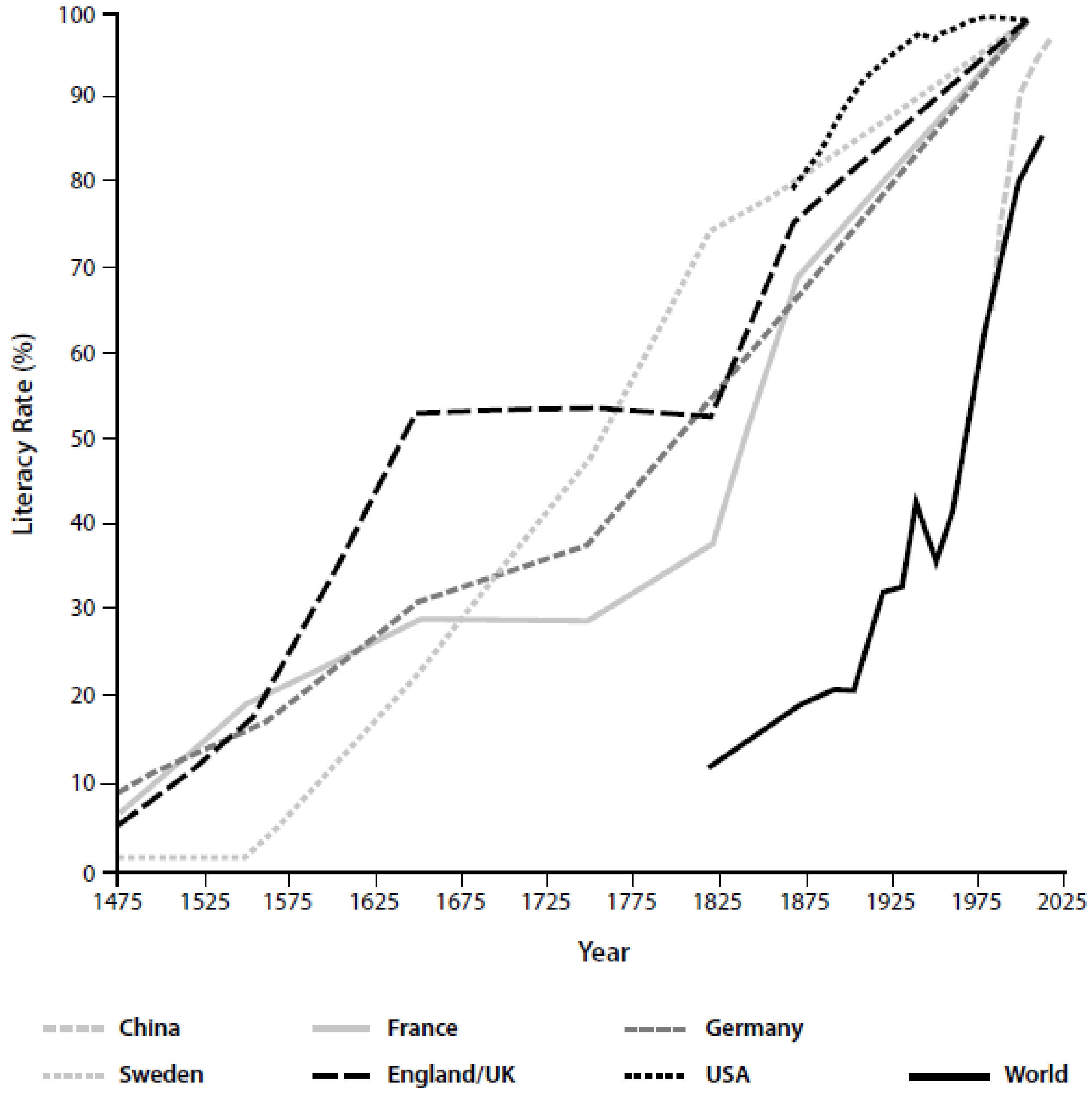


Figure 7. Rising Literacy Rates around the World, 1475–2010[12]

Addie Card, 12 Years. Spinner in North Pownal Cotton Mill. Vermont, 1910[41]

Tractor Advertisement, 1921

Keep The Boy In School

THE pressure of urgent spring work is often the cause of keeping the boy out of school for several months. It may seem necessary – but it isn't fair to the boy! You are placing a life handicap in his path if you deprive him of education. In this age, education is becoming more and more essential for success and prestige in all walks of life, including farming.

With the help of a Case Kerosene Tractor it is possible for one man to do more work in a given time, than a good man and an industrious boy, together, working with horses. By investing in a Case Tractor and Grand Detour Plow and Harrow outfit now, your boy can get his schooling without interruption, and the Spring work will not suffer by his absence.

Keep the boy in school – and let a Case Kerosene Tractor take his place in the field. You'll never regret either investment.[47]

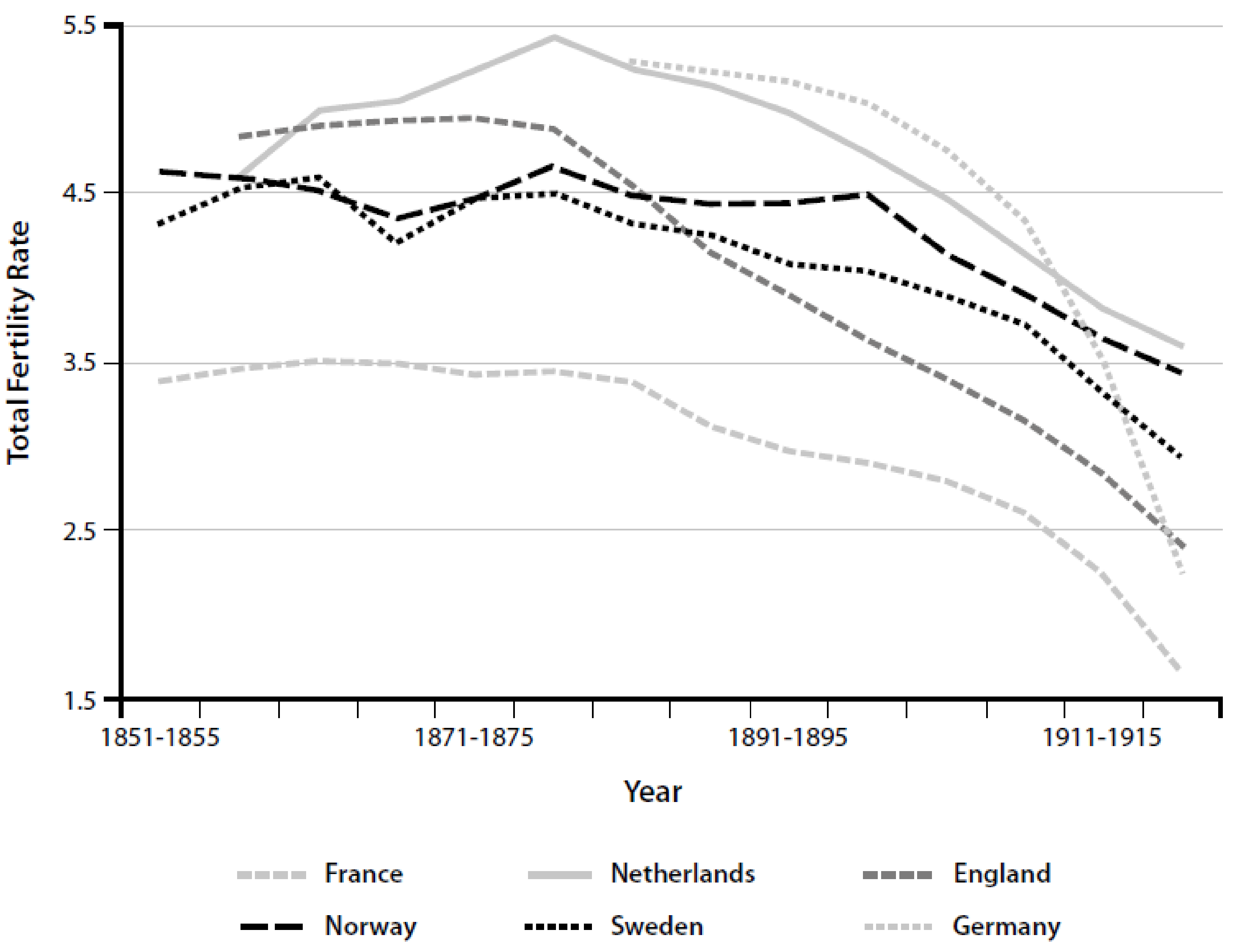


Figure 8. Children per Woman

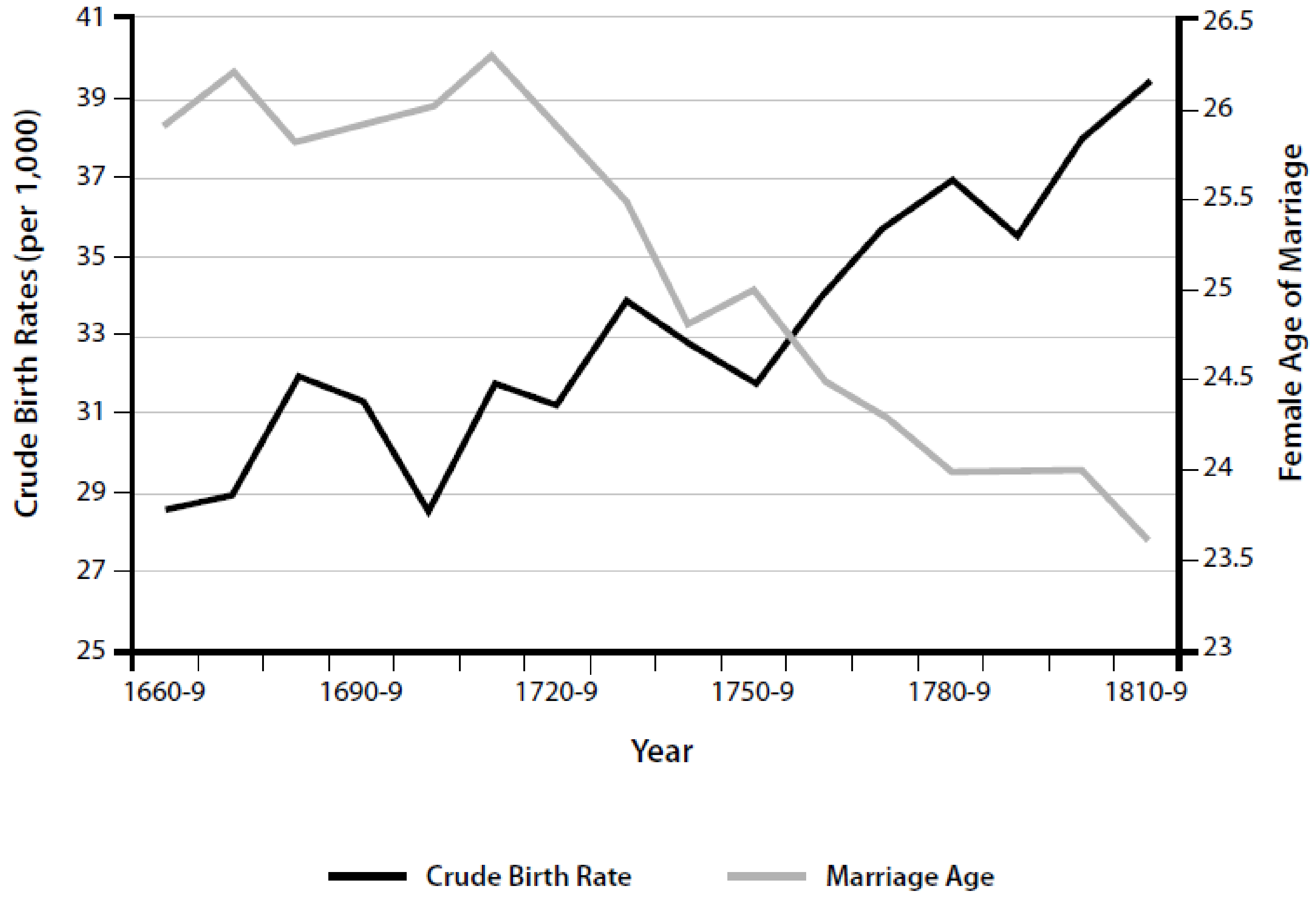
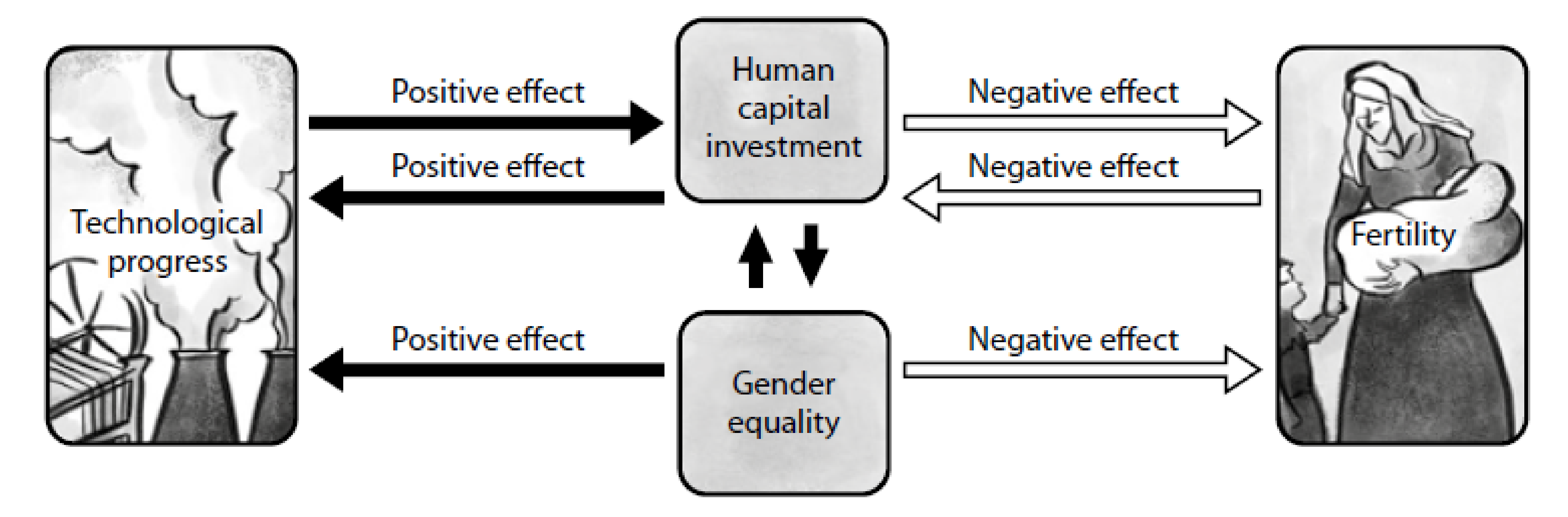
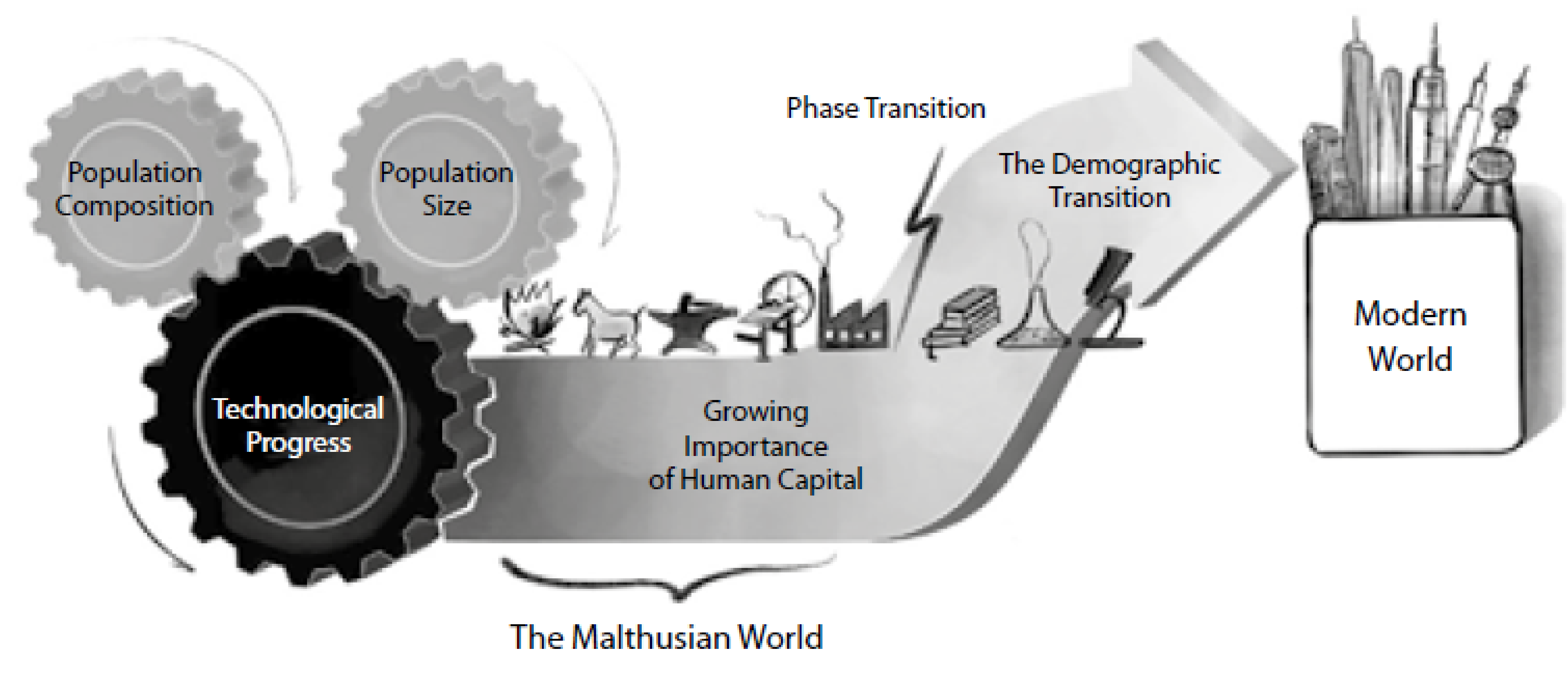


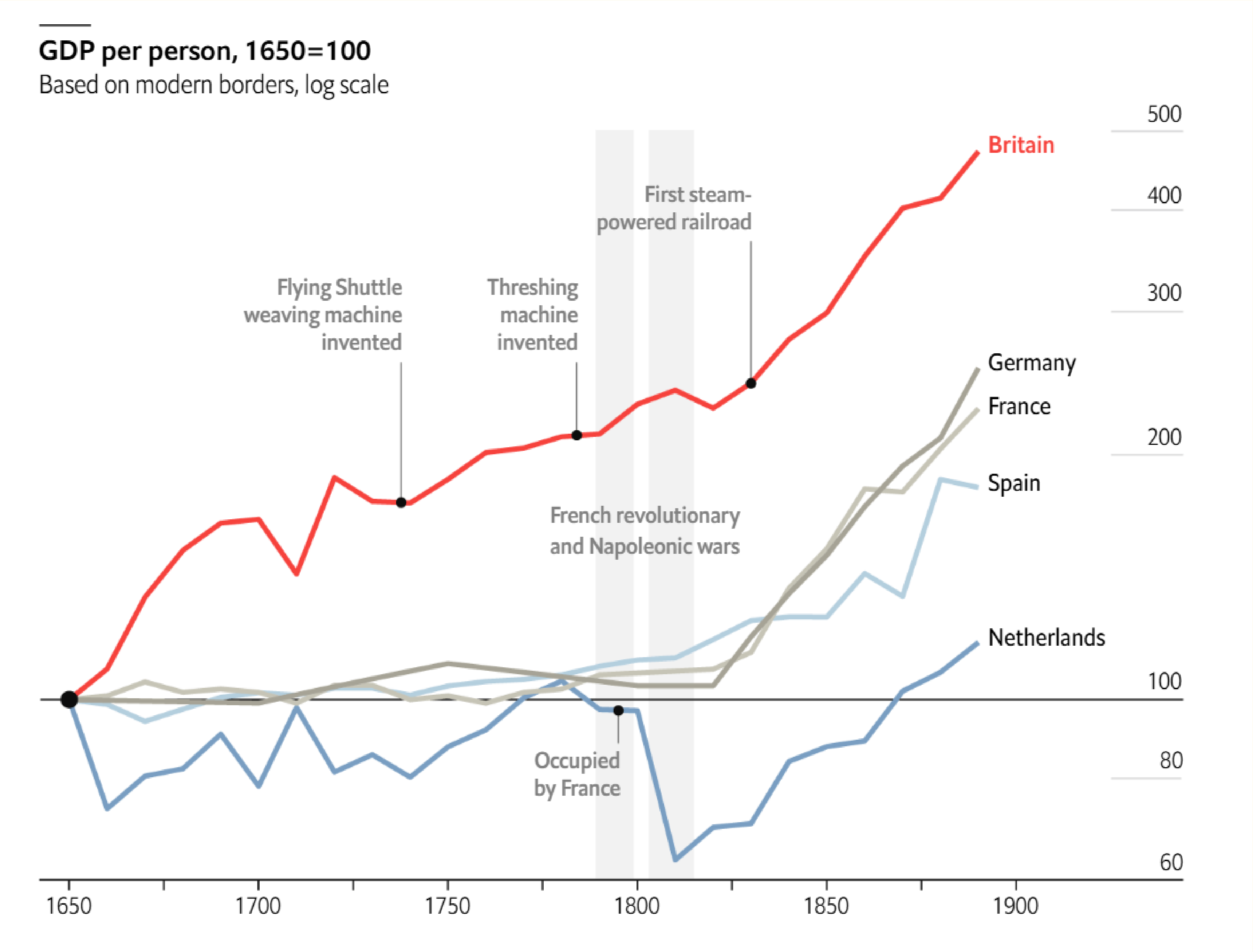
Figure 9. Fertility Rates and Women's Age of Marriage in England, 1660–1820

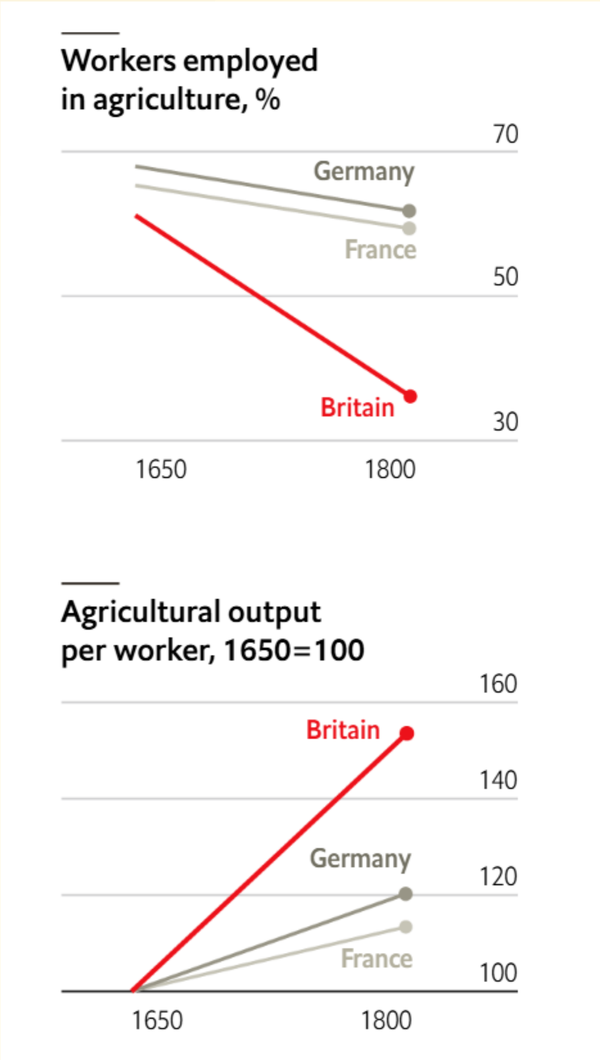


Technological progress
Positive effect
Positive effect
Human capital investment
Negative effect
Negative effect
Fertility
Positive effect
Gender equality
Negative effect



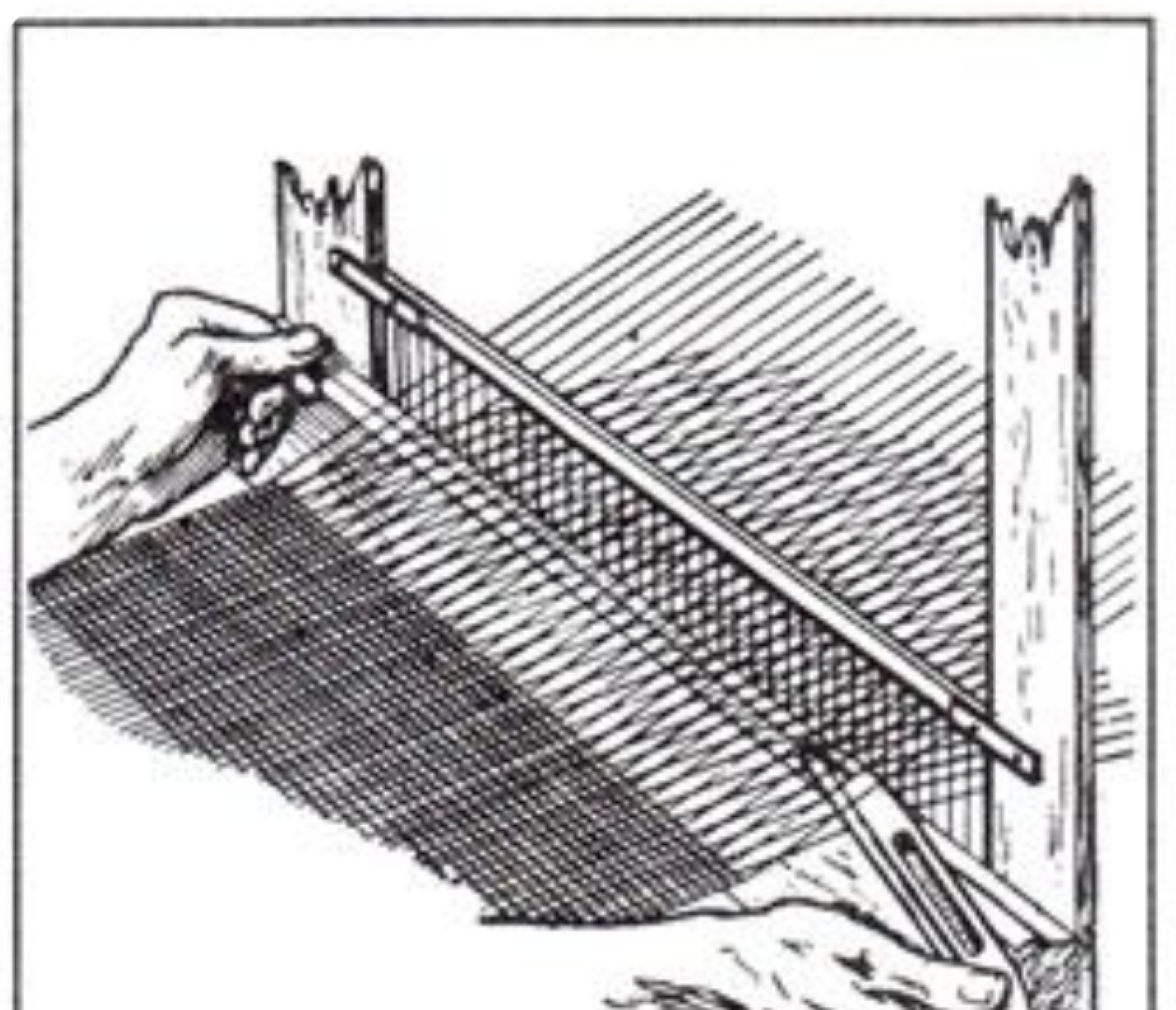
The Cogs of Change


GDP per person, 1650=100
Based on modern borders, log scale
500
400
300
200
100
80
60
Britain
Germany
France
Spain
Netherlands
Flying Shuttle weaving machine invented
Threshing machine invented
First steam-powered railroad
French revolutionary and Napoleonic wars
Occupied by France
1650
1700
1750
1800
1850
1900


Workers employed in agriculture, %
70
Germany
France
50
Britain
30
1650
1800
Agricultural output per worker, 1650=100
160
Britain
140
Germany
120
France
100
1650
1800

Textile innovations

- John Kay (1733): Flying shuttle. Increased speed of weaving.
- James Heargreaves (1764): Spinning Jenny. Allowed multiple spindles to be spun simultaneously.
- Richard Arkwright (1769): Water frame. Machine powered machines, pivotal to move production from homes to factories. For strong warp threads. Also developed carding machines for processing raw cotton.
- Samuel Crompton (1779): Spinning mule. Combined spinning jenny and water frame to produce fine strong yarn.



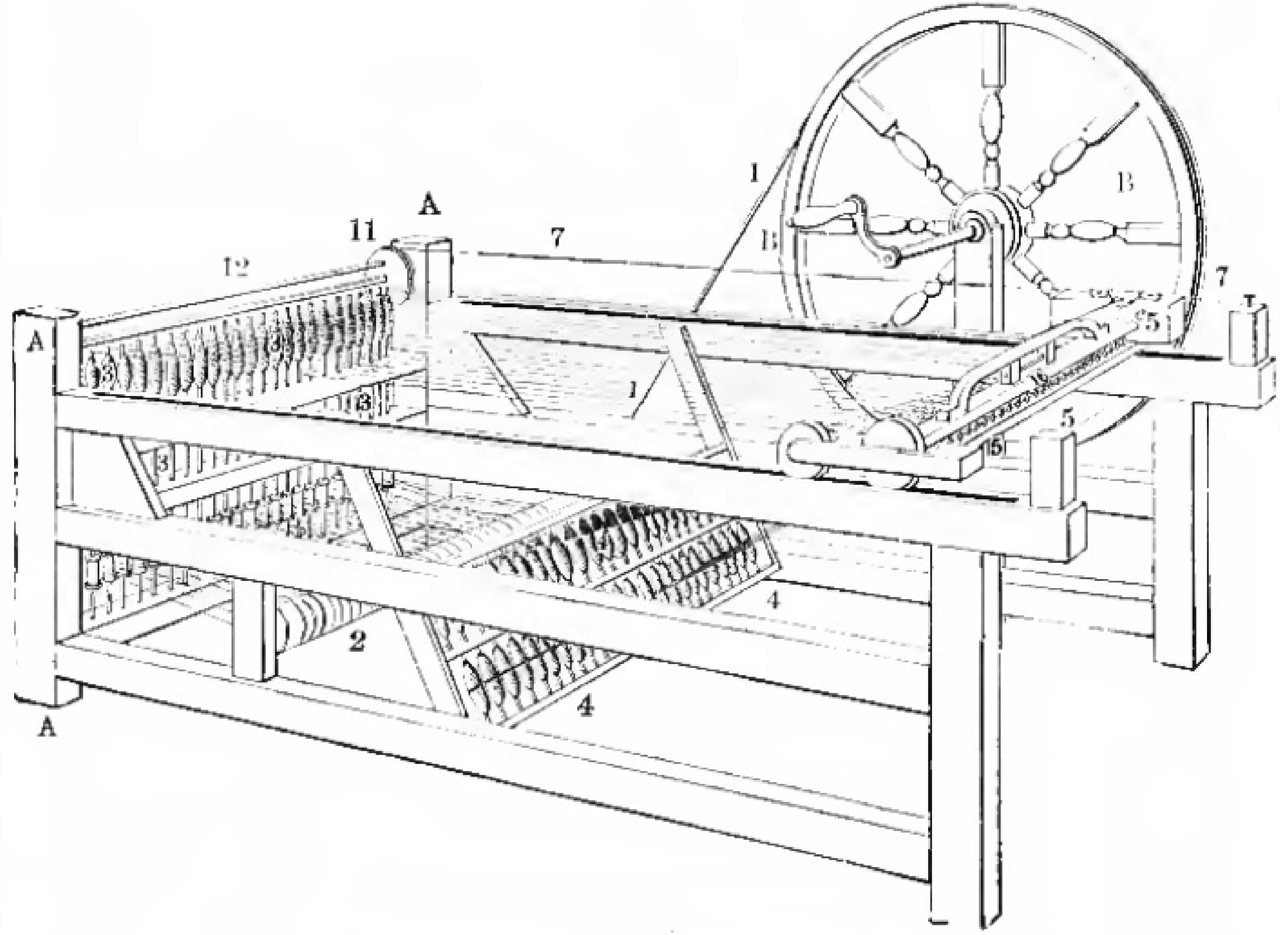


Figure 8.3 The spinning jenny

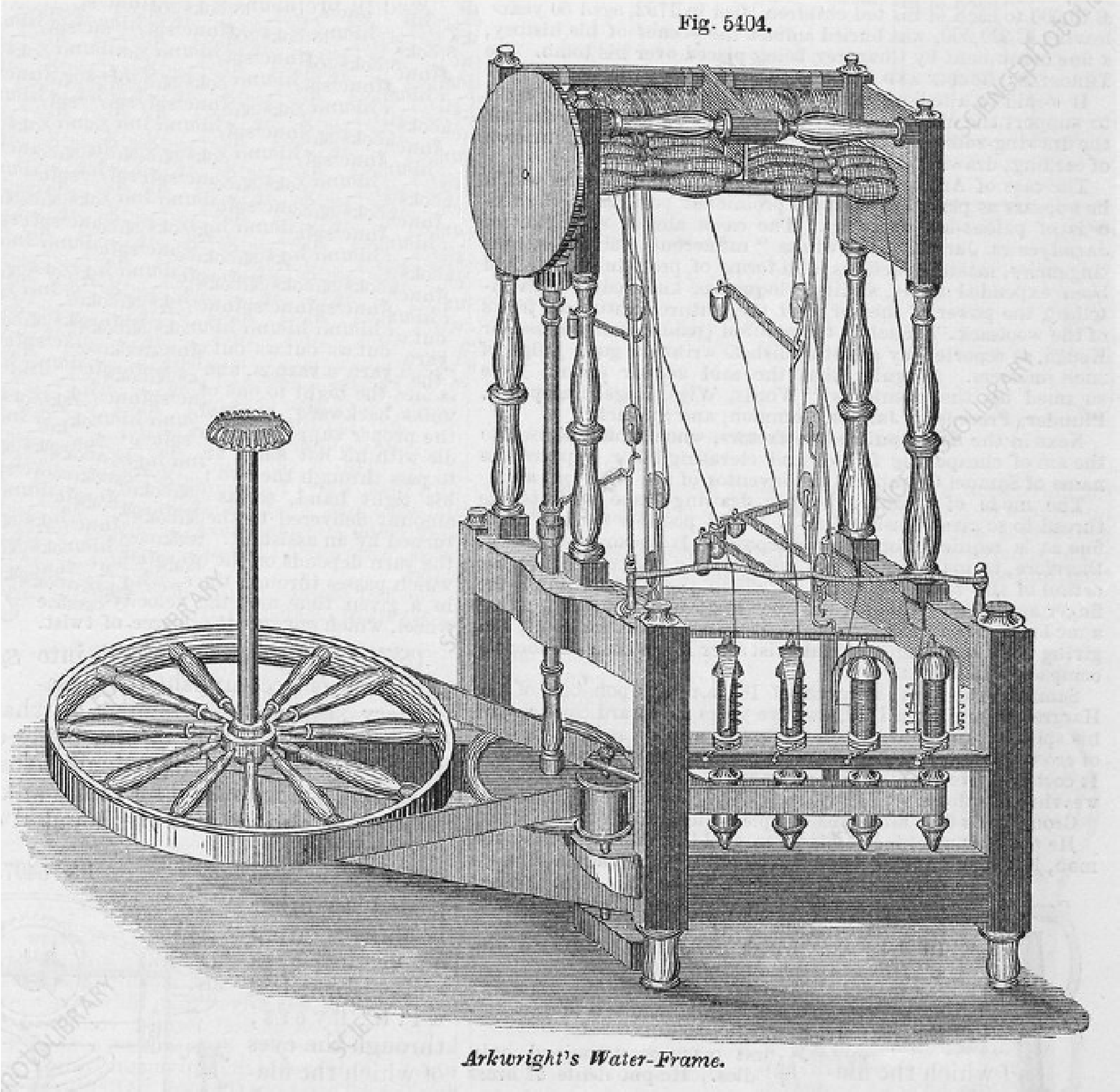


Figure 8.4 Arkwright's water frame

Arkwright's Waterframe, patented 1769

https://youtu.be/AloWMoc-3WU?si=uug7Berx47RHEQHQ

Spinning Mule (1779)

- Faster, higher quality

longer: https://youtu.be/aBRLhqoc6yE?si=GhPIp_NwkSsvxVV8&t=266

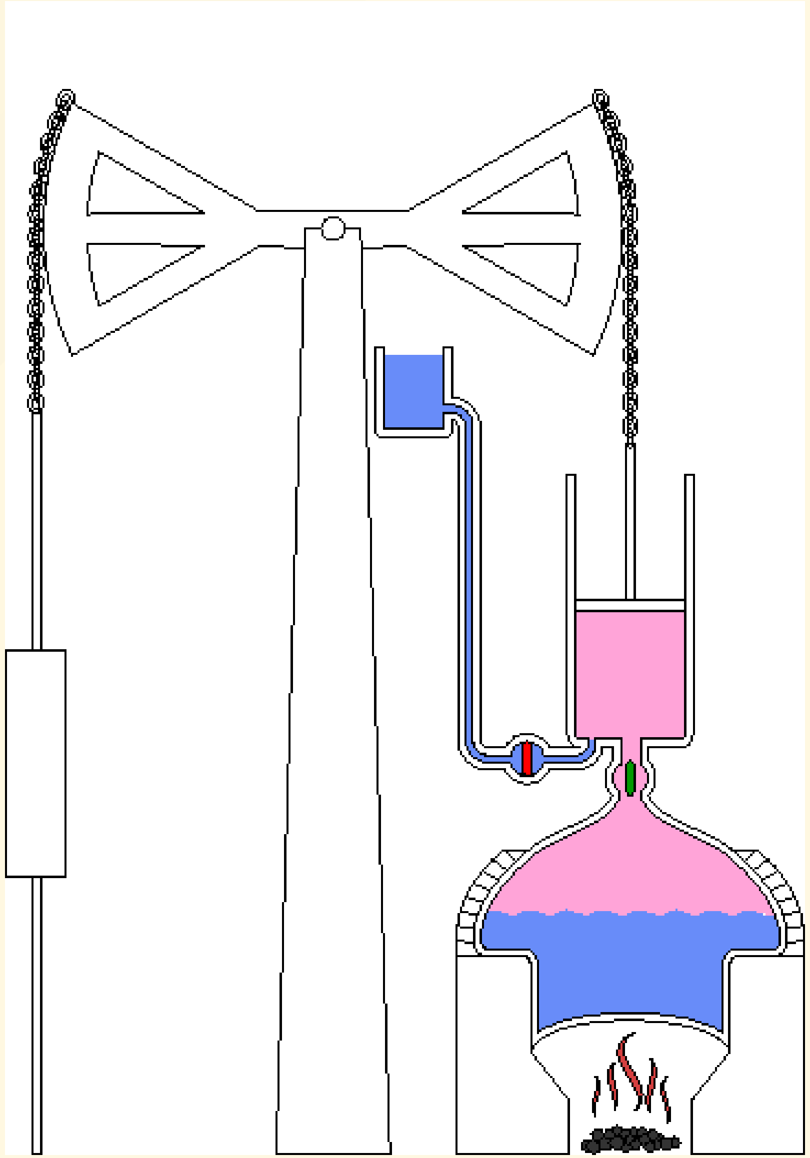
Mokyr: Macro vs. Microinventions

- **Macroinventions:** Radical, unprecedented leaps in technology that break major bottlenecks (e.g., discovering atmospheric pressure can do work).
- **Microinventions:** Continuous, incremental improvements made by skilled mechanics and engineers that make macroinventions reliable and economically viable.
- *The British Industrial Revolution relied heavily on a culture that fostered both.*

Atmospheric Engine (A Macroinvention)

- Newcomen's 1712 engine: A radical conceptual leap, but highly inefficient.
- Water spray cools steam in chamber, causing condensation.
- Atmospheric pressure pushes piston down, lifting weights.

See [animation]



James Watt's Steam Engine (Macro & Microinventions)

- Watt 1769 design, greatly improves energy efficiency
  - Separate condensing cylinder (Brilliant macroinvention)
  - Avoided need to reheat chamber
  - Required years of collaborative microinventions (with Matthew Boulton and precision ironmaker John Wilkinson) to mass produce.

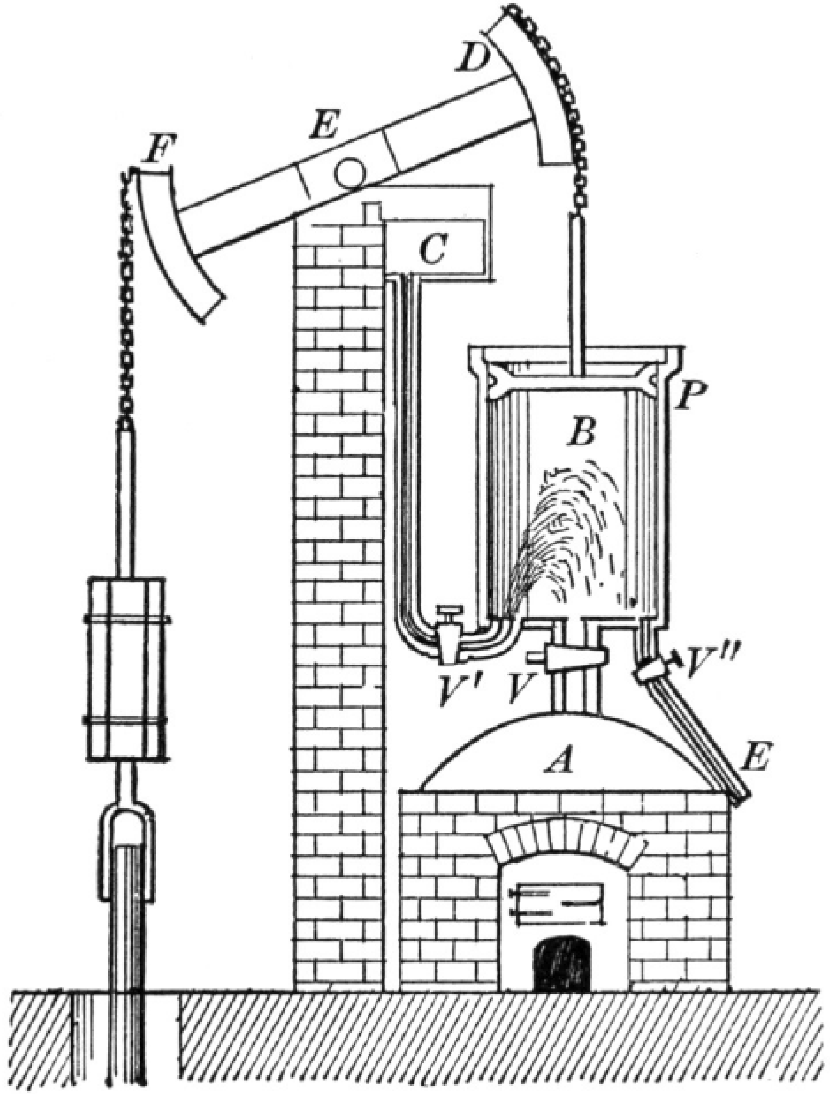
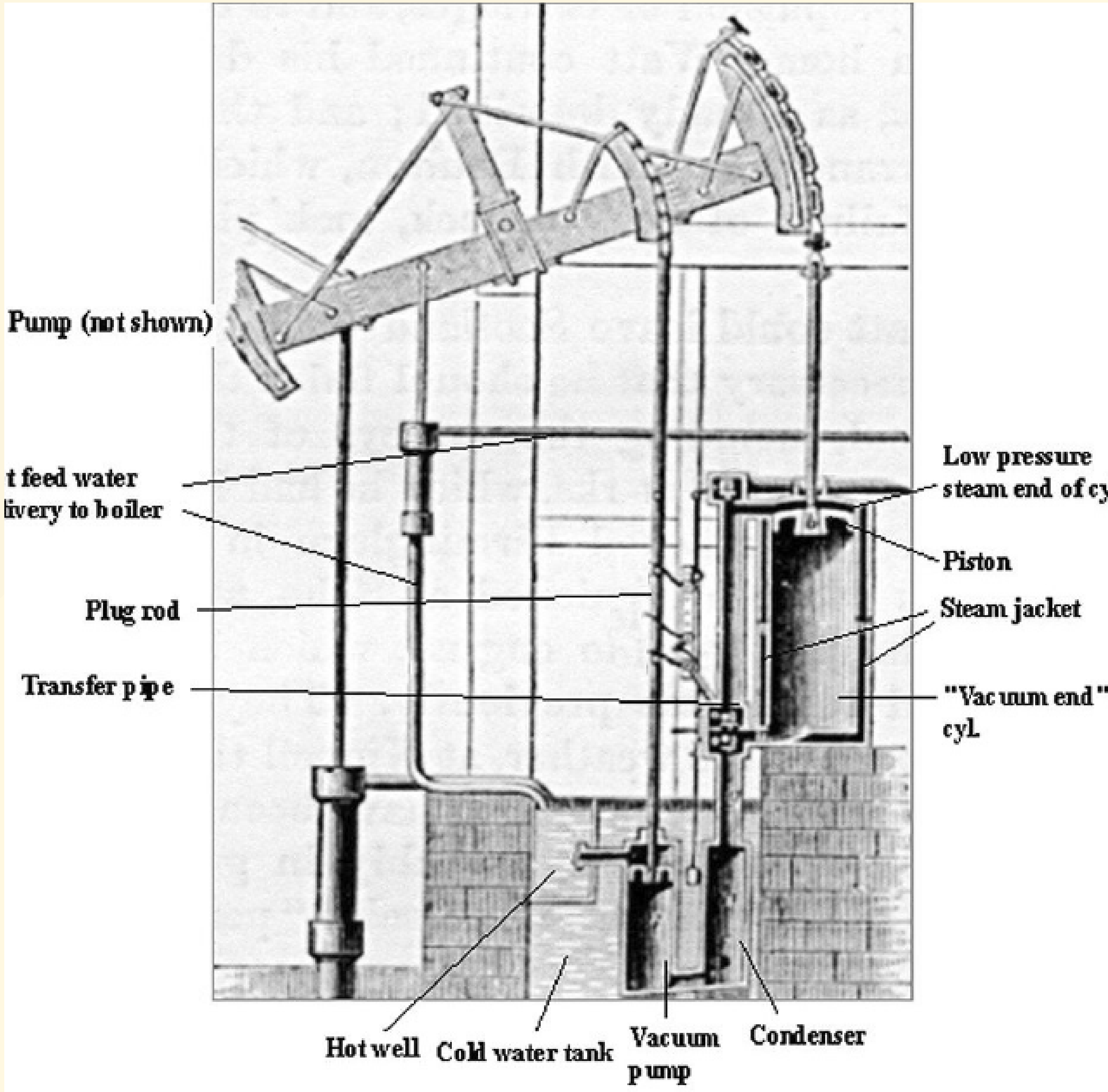


Figure 8.6 Schematic of the Newcomen steam engine



ure 8.7 The Watt steam engine

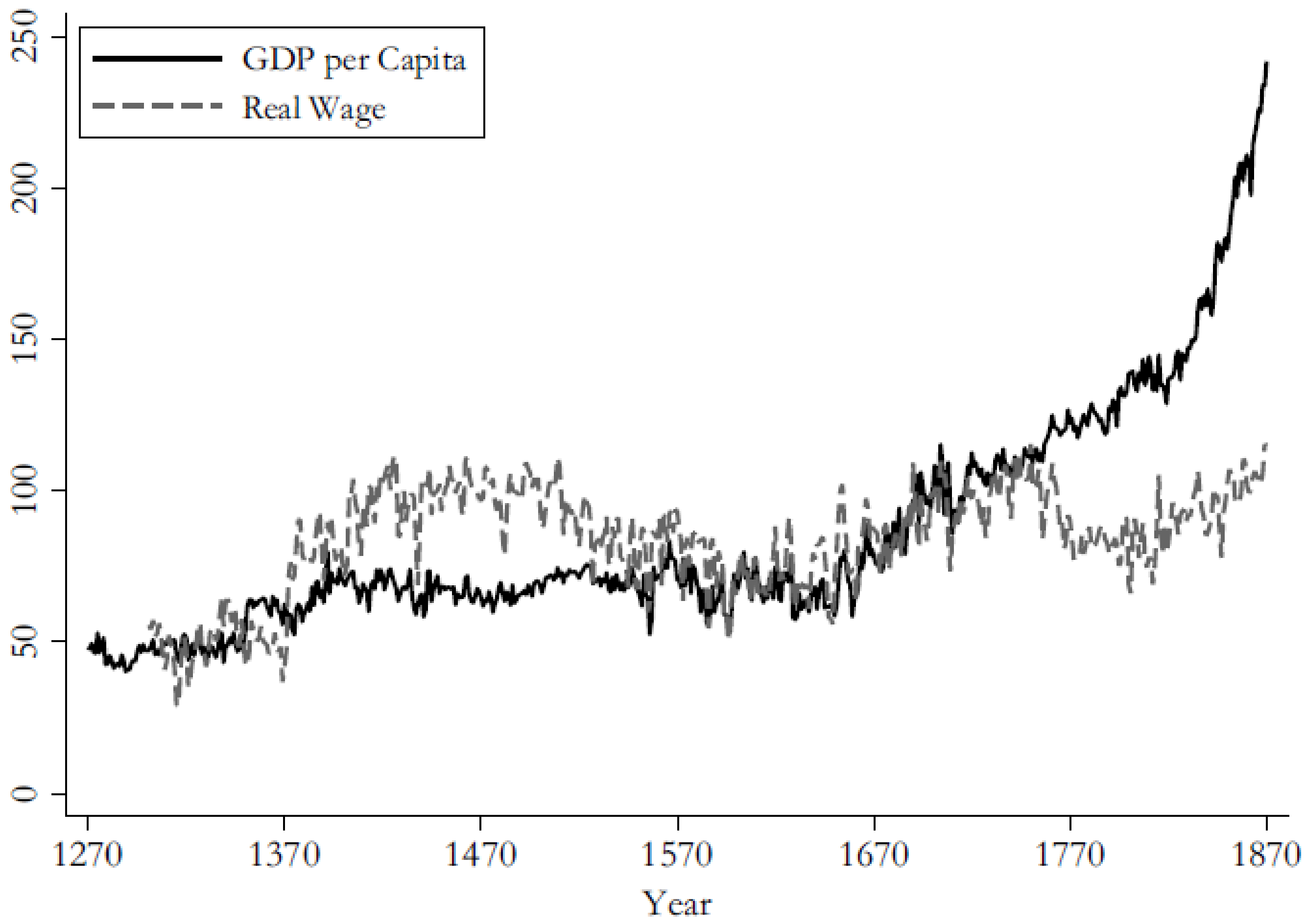


Figure 9.1 GDP per capita and real wages in England/Great Britain, 1270–1870 (1700 = 100)

Data sources: Broadberry, Campbell, Klein, Overton, and van Leeuwen (2015) for GDP data, Allen (2001) for real wage data. Data refer to England between 1270 and 1700 and Great Britain from 1700 to 1870.

The Geography of British Industrialization

- Rise of London part of broader shift towards ports, especially those involved in Atlantic trade
- Bristol and Liverpool grew on trade with the Americas, especially the slave trade
- Industrialization shifted economic activity to the northwest
- New economic centers in Manchester, Leeds, Birmingham and Sheffield based on manufacturing, textiles, and steel.

Table 8.1 Most populous English cities excluding London (with populations of at least 10,000)

1520	**1600**	**1700**	**1801**
Norwich	Norwich	Norwich	Manchester
Bristol	York	Bristol	Liverpool
York	Bristol	Newcastle	Birmingham
	Newcastle	Exeter	Bristol
		York	Leeds
		Yarmouth	Sheffield
			Norwich
			Portsmouth
			Bath

Data source: Clark (2001).

Why Britain? KRs Chapter 8

List of preconditions:

- Relatively 'limited' (yet strong) and representative governance.
- Large domestic economy
- Access/control of Atlantic trade
- Large base of highly skilled mechanical workers

Other Contenders include

- Slavery and Colonization <-- discussed next class.

Two Dominant Theories

- Allen: high wages and relatively cheap energy
- Mokyr: 'Industrial Enlightenment' Science + mechanical skill

Mokyr's Culture of Growth

- Individual scientific genius not enough (plenty outside of Britain)
- Scientific societies like the Royal Society (founded in 1660)
- Academic journals
- reliable Europe-wide postal system
- Comparative intellectual freedom
- A market for ideas -- - Voltaire's Republic of Letters.
- Culture not just academic; influenced businessmen, mechanics and engineers.

To understand Allen's theory we need to understand

Technology Adoption and directed technical change

See interactives: https://www.geogebra.org/m/hzy5yyeg#chapter/927950

A Cobb-Douglas Aggregate production function:

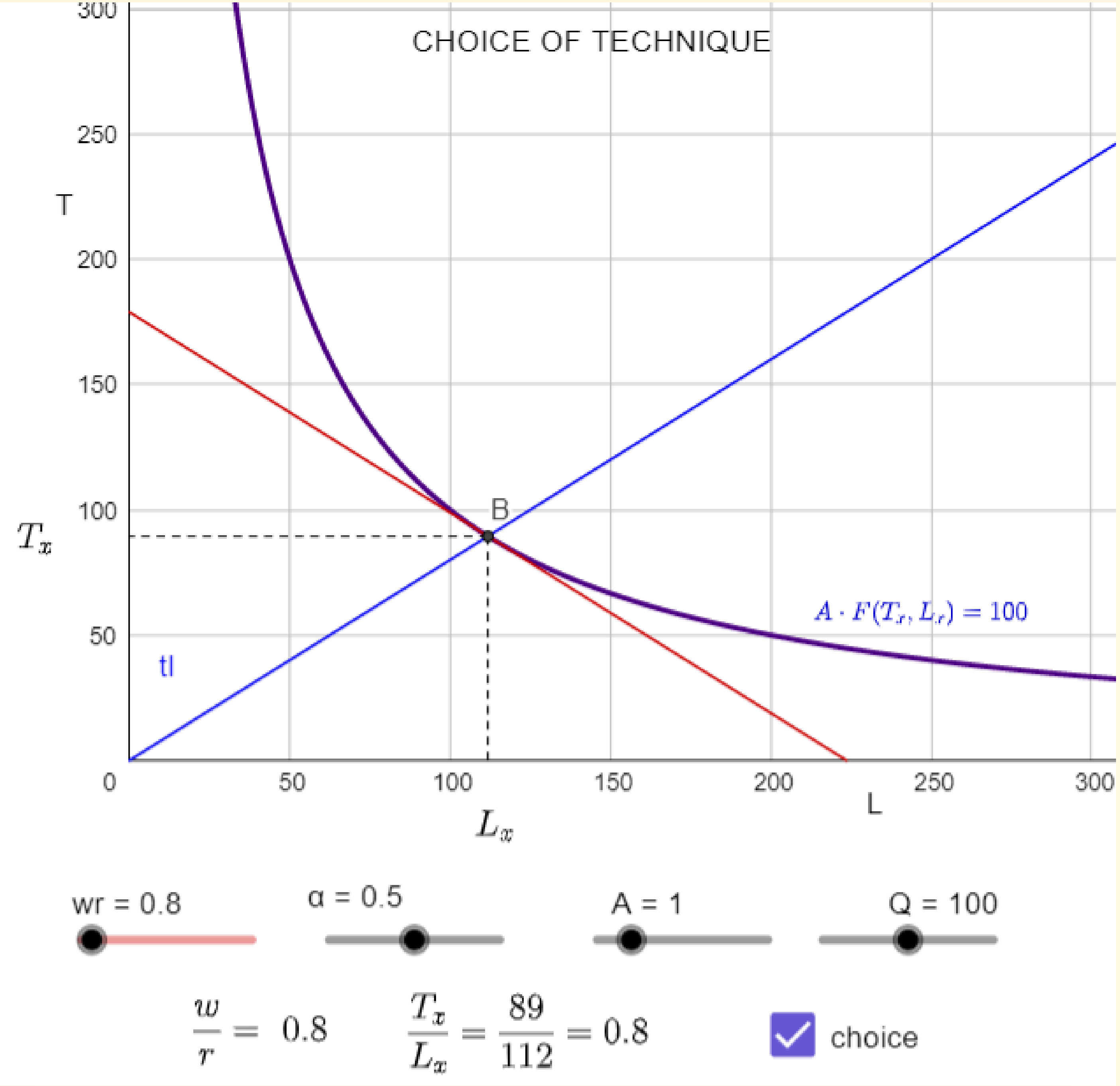
$$Y = A \cdot K^{\alpha} \cdot (H \cdot L)^{1-\alpha}$$

With human-capital augmented laborL

$$Y = A \cdot K^{\alpha} \cdot (H \cdot L)^{1-\alpha}$$

Optimum Choice of Technique depends on economy w/r

interactive



Adoption incentives

- Allen (2009) argues that regardless of how intensive the machine was being used, it was always profitable for industrialists in England to purchase a spinning jenny.
- Often not profitable for a French industrialist to do so (French wages were lower and capital more expensive)
- Never paid to adopt a spinning jenny in India.

TABLE 1
RATES OF RETURN TO BUYING A SPINNING JENNY IN BRITAIN, FRANCE, AND INDIA

Relative Productivity	Percent Full-Time	Britain (percent)	France (percent)	India (percent)
2	0.5	34.6	0.2	–7.7
2	0.4	24.0	–8.2	–17.4
2	0.3	12.3	–21.7	–100.0
3	0.5	51.2	10.7	3.0
3	0.4	38.0	2.5	–5.2
3	0.3	24.0	–8.2	–17.3
4	0.5	59.2	15.3	7.3
4	0.4	44.7	6.8	–0.1
4	0.3	29.4	–3.7	–12.0

Note: The relative productivity refers to labor productivity with a jenny relative to a spinning wheel.
Sources: See the text.

Slow diffusion of power looms

- Spinning mechanized before weaving (looms).
- A lot of cheap yarn lead to weaving boom 1780s-1815. Cotton *hand* weavers 4X as farmers took up weaving full-time.
- ~250K hand looms in 1823 better at high quality cloth.
- For highest quality cloths, wage tripled (Allen 2016).
- Allen high weaver wages induced invention and adoption of labor-saving power looms
- Tariffs restricted India textiles.

Table 3. Number of power looms in Britain 1803–57[68]

Year	1803	1820	1829	1833	1857
Number of looms					
in England		12,150	45,500	85,000	
in Scotland		2,500	10,000	15,000	
Total	2,400	14,650	55,500	100,000	250,000

[Source: Hills (1989), pg. 117. In 1823, there were 240,000 to 250,000 hand looms according to Ray (2009).]

Rising K/L and labor productivity (source)

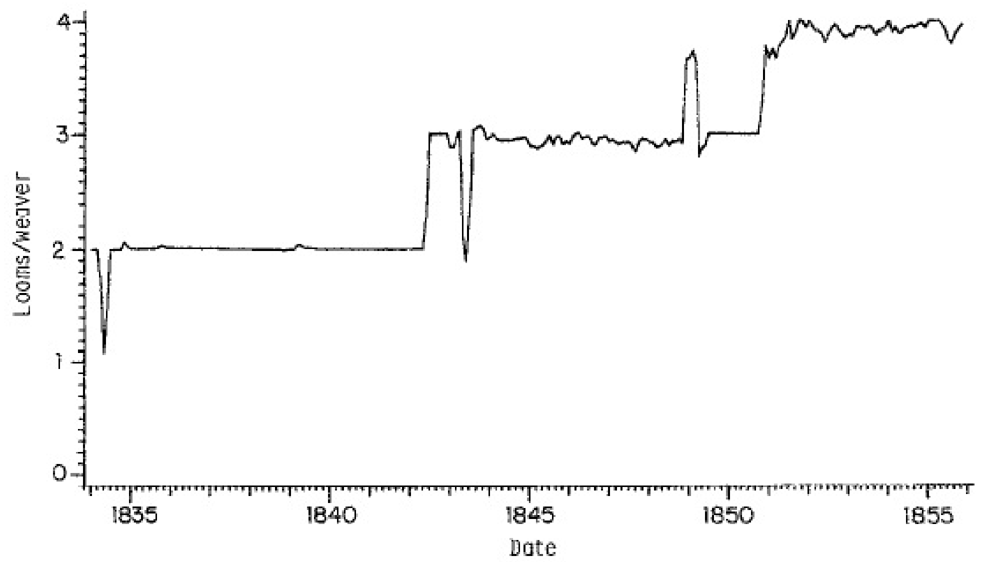


FIG. 4. Average number of looms per weaver, January 1834–November 1855. No data available for May 1836 through March 1838. For variable definition, see Appendix.

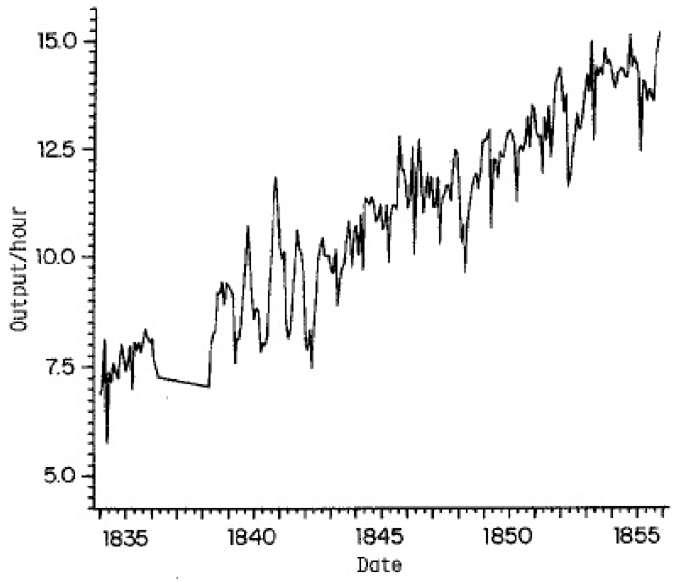


FIG. 1. Output per weaver–hour, January 1834–November 1855. (Yards of "C" cloth.) No data available for May 1836 through March 1838. For variable definition, see Appendix.

Slow diffusion of Steam

- Steam engines used in mines, pumps, mills, locomotives.

Table 3

Sources of Power, 1760–1907 (Horsepower)

	1760	1800	1830	1870	1907
Steam	5,000	35,000	165,000	2,060,000	9,659,000
Water	70,000	120,000	165,000	230,000	178,000
Wind	10,000	15,000	20,000	10,000	5,000
Total	85,000	170,000	350,000	2,300,000	9,842,000

Source: Kanefsky (1979*a*, p. 338); not including internal combustion engines.

Steady productivity growth

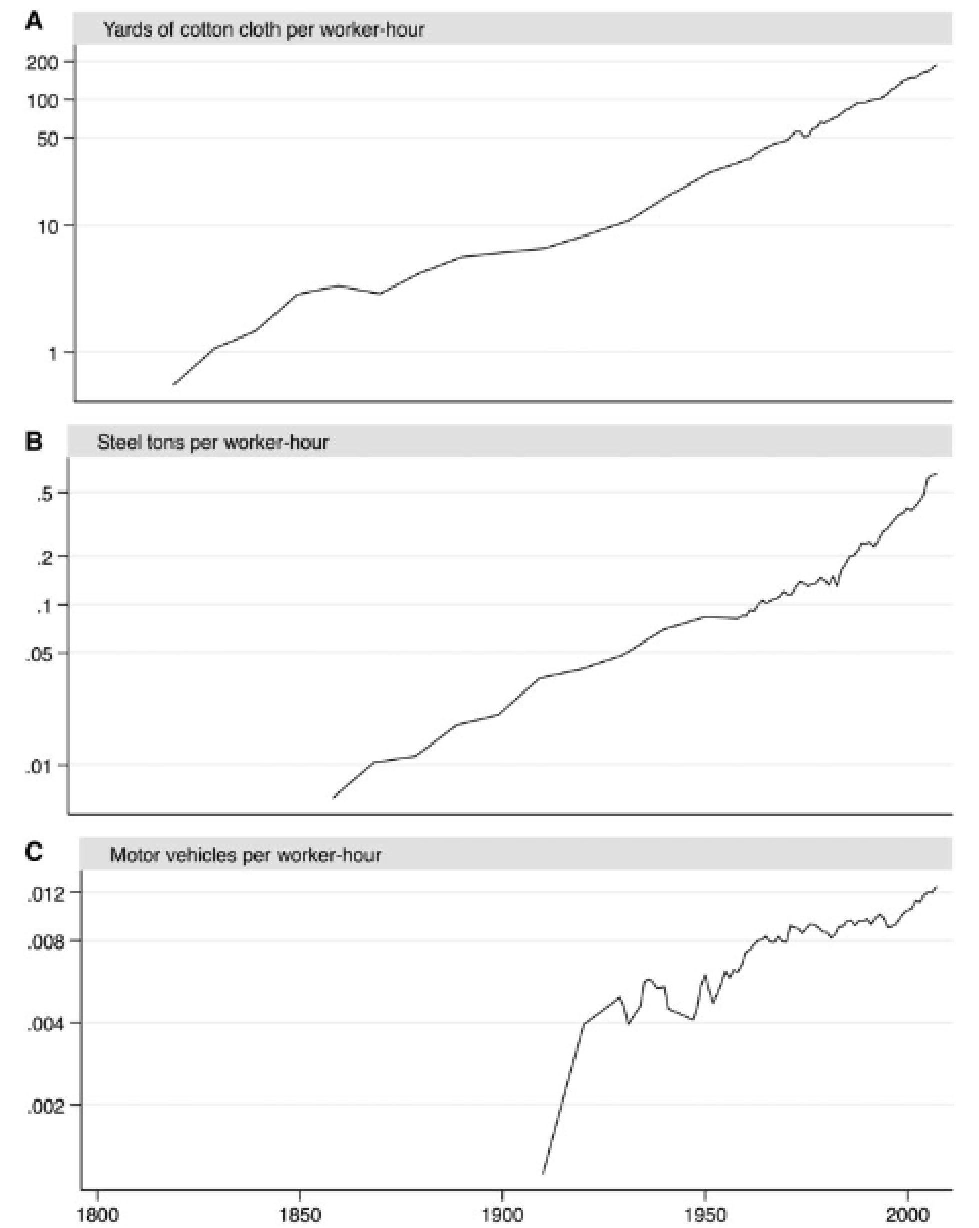


Figure 2. Labour productivity over time. (A) Yards of cotton cloth per worker-hour, (B) steel tons per worker-hour and (C) motor vehicles per worker-hour.

Does productivity growth increase or decrease employment?

- Firm hires laborers until marginal value product of labor equals the wage.

$$P \cdot MP_L = w$$

- At constant price P firm would want to hire more workers as productivity increases.
- Many firms doing the same pushes up wages.
- But what is happening to the demand for the good, and it's price P? It may fall if supply rising faster

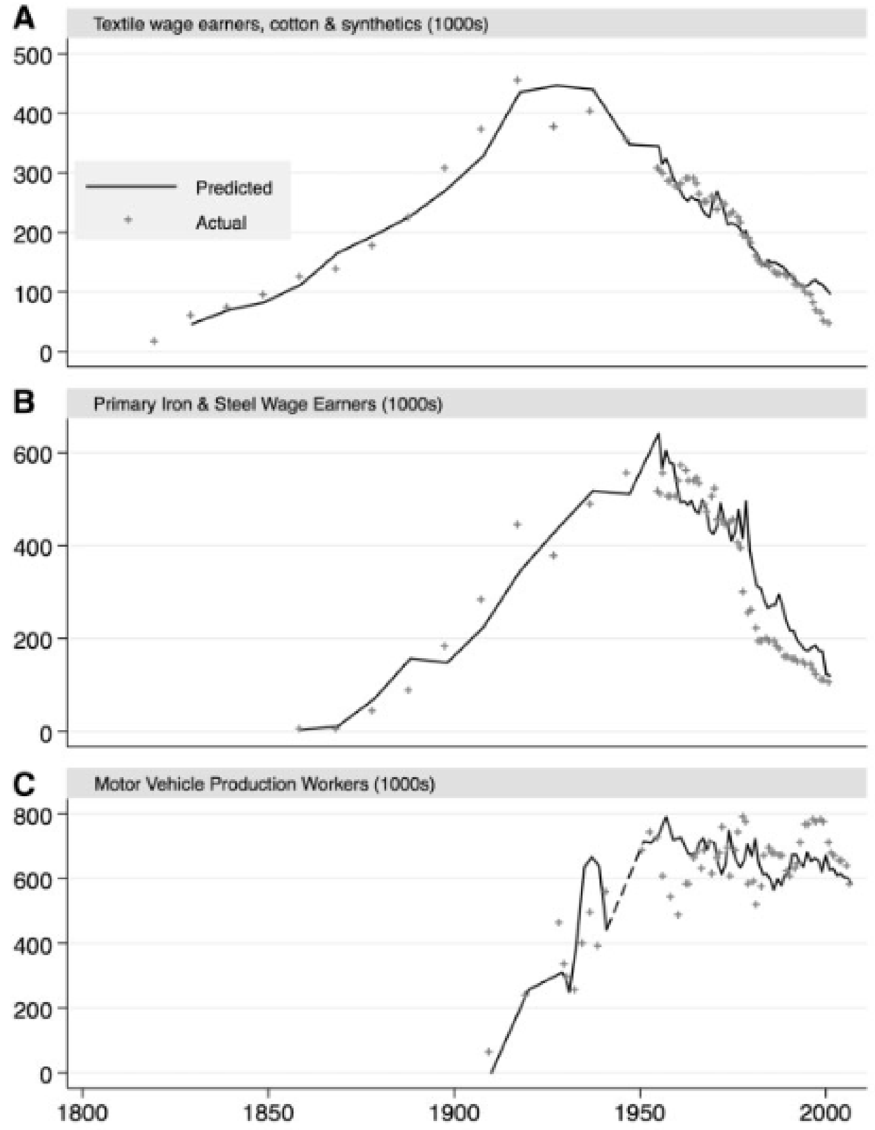


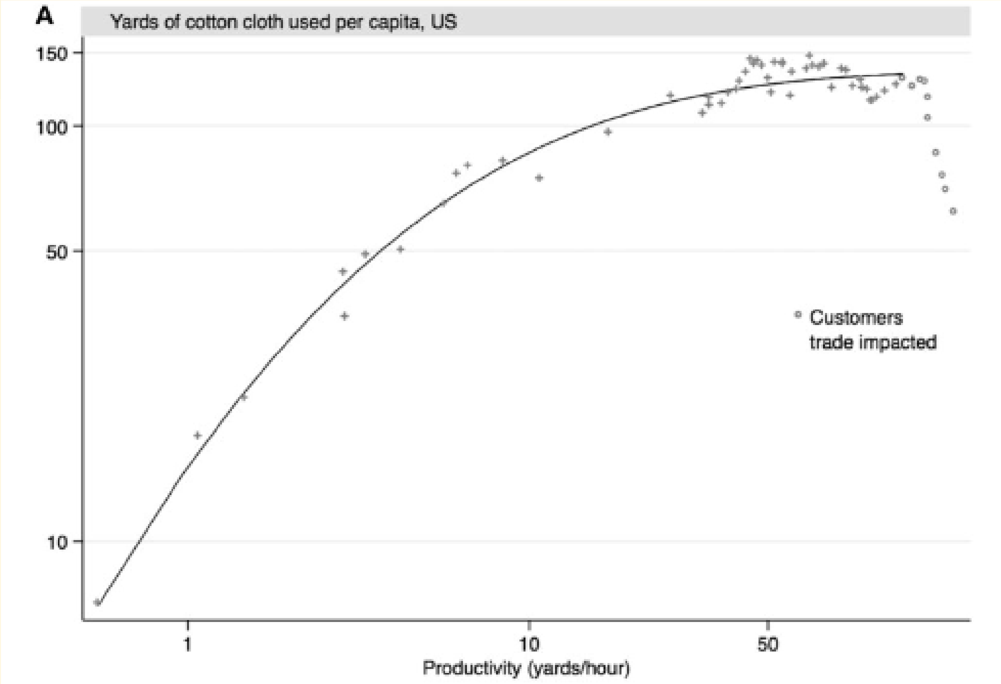
Figure 1. Production employment in three industries. (A) Textile wage earners, cotton and synthetics (1000s), (B) primary iron and steel wage earners (1000s) and (C) motor vehicle production workers (1000s).

Employment and wage dynamics

- Demand for cloth first grew quickly. Price P rose as demand rose faster than supply, so labor demand and employment increases (wage may rise if supply inelastic) :

$$P \cdot MP_L = w$$

- But eventually demand growth slows and product price P falls faster than productivity rises. Supply outstrips demand.
- Industry would now start shedding workers as fewer (more productive workers) required to meet demand.

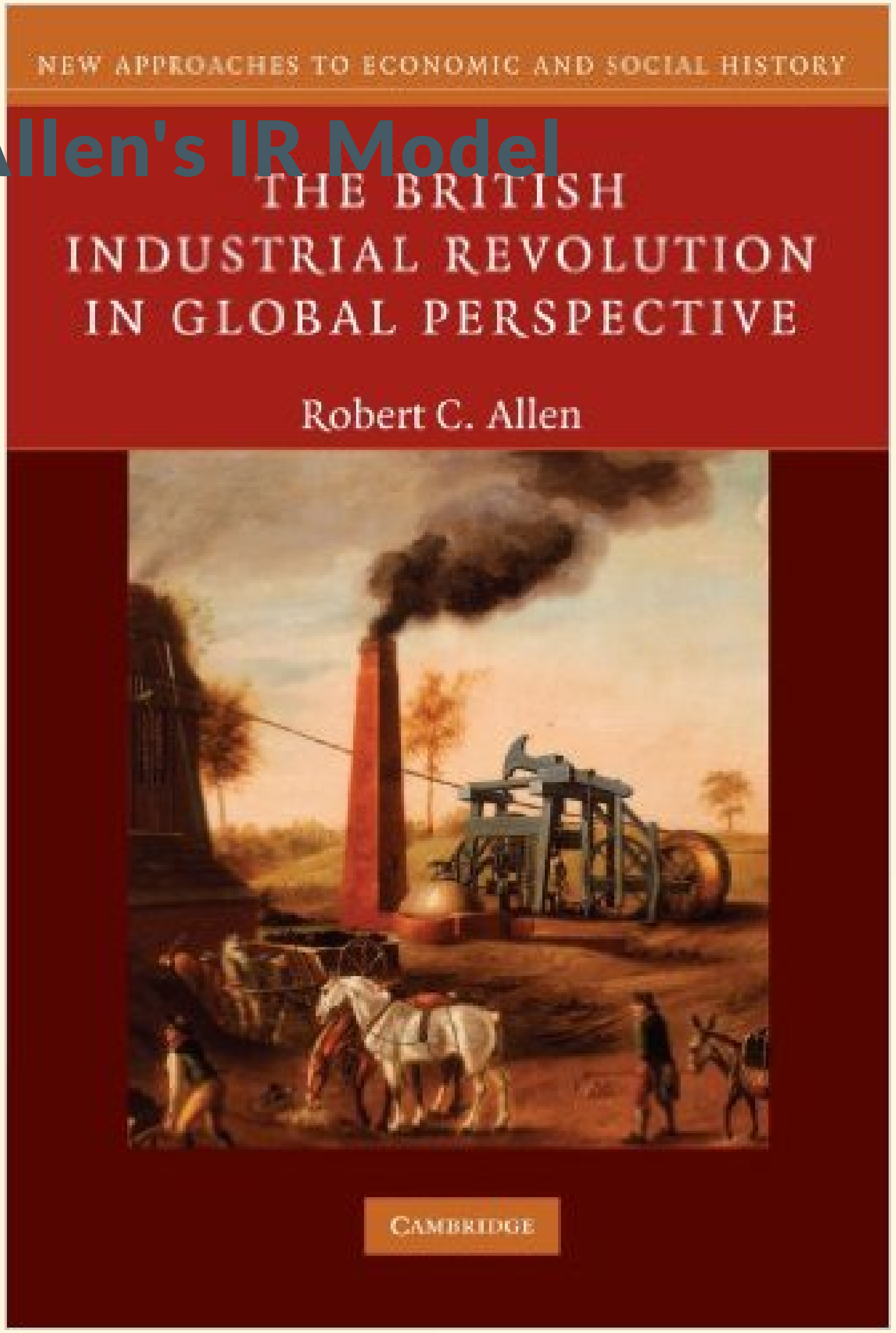


Microeconomic mechanisms

Geogebra interactives

- MPL and K/L
- Optimum choice of K/L technique (given w/r)

Allen's IR Model



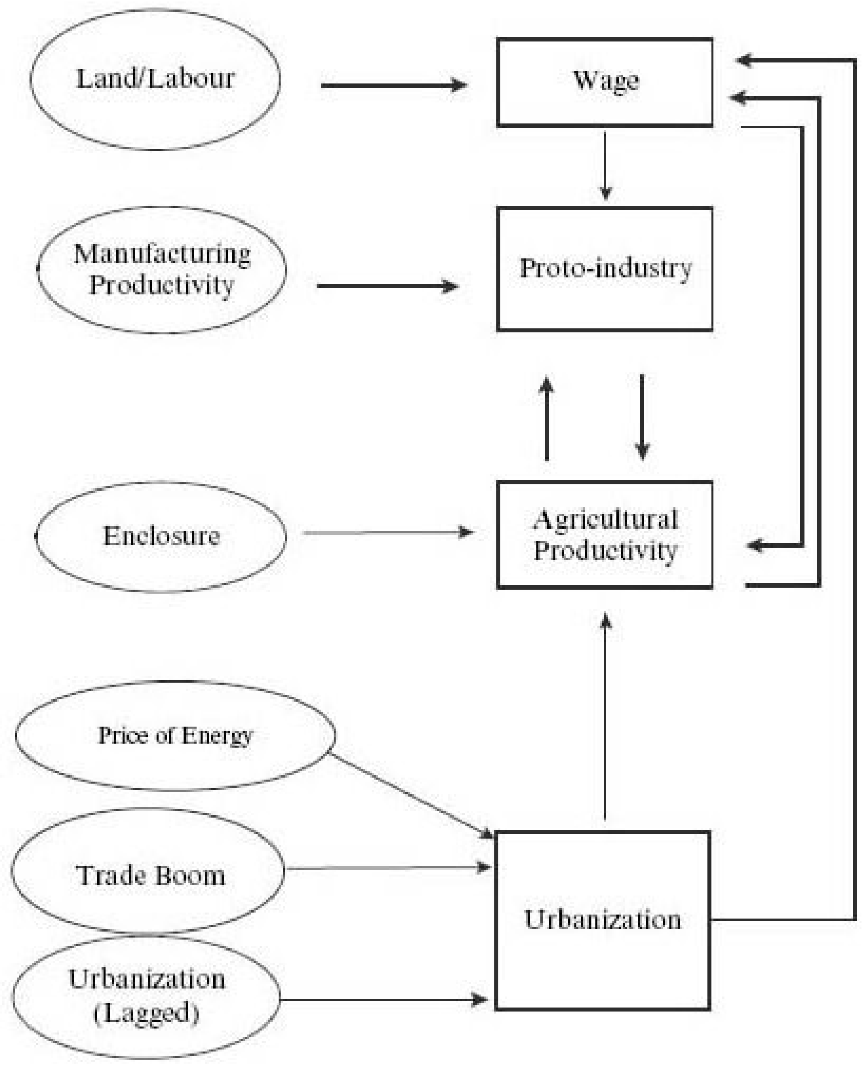


Figure 5.1 Flowchart (one period) of the model